Weight-Space Anatomy and Zero-Training LoRA Surgery: Breaking Giant-Bucket Over-Merging Beyond On-Policy Self-Distillation

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Abstract

When post-training Small Language Models (SLMs) such as `Qwen/Qwen3.5-4B` for complex multi-task Browser Tab Grouping (`v2.1`), standard Supervised Fine-Tuning (`sft2000`) suffers from severe *giant-bucket over-merging*—collapsing 20–25 heterogeneous tabs into a single catch-all group (45.71% Recall, 58.33% Initial Precision). Distilling from 1,200 frontier (`GPT-6-astra`) reference solutions via On-Policy Self-Distillation (`opsd_astra`) or off-policy SFT (`sft_astra`) lifts `v2.1` Recall to 59.36% and 60.25%, respectively, yet each paradigm remains bottlenecked: OPSD's lazy trajectory gradients under-update middle-layer feature splitters, whereas off-policy SFT corrupts late readout layers (65.08% Precision vs. OPSD's 69.09%).

Inspired by *Weight-Space Programming* (Takehi et al., 2026), we perform an exact 32-layer QR-SVD and unembedding-projection anatomy of the 21.2M-parameter LoRA updates $\Delta W = W - W_0$ without running gradients. We discover an exact functional boundary at **Layer 18**: (1) middle layers (**L08–L15**) form a shared cross-tab clustering circuit ($\cos(\Delta W_{\text{OPSD}}, \Delta W_{\text{SFT}})$ up to 0.41) where SFT's update norm is 1.6× stronger, whereas (2) late layers (**L18–L31**) diverge to near-orthogonality (cos down to 0.01), where SFT's leading singular directions project 8×–11× more strongly onto surface lexical artifacts (`' &'`) and aggressive `Ungrouped` thresholds. Directly guided by this map, we establish a clean monotonic ladder of **zero-training weight-space surgeries** on `v2.1`: (i) *Single-Model Neuron Surgery* (**Idea A**: scaling 64 newline-vs-comma MLP columns per layer in L26–L31 of base `sft2000`) jumps `v2.1` Recall from 45.71% to 57.93% **(+12.22pp)** with zero training; and (ii) *Dual-Adapter Circuit Grafting & Spectral Consensus* (`Graft-18` and `SVD-Blend-50/50`) strictly dominate both OPSD and SFT on `v2.1`, pushing Recall to 64.52% **(+5.16pp over OPSD)** and Initial Precision to 72.41% **(+3.32pp over OPSD, +7.33pp over SFT)**.

1 Introduction

Post-training compact language models (2B–4B parameters) for structured clustering and multi-entity organization requires balancing two distinct capabilities: *internal relational representation* (distinguishing fine-grained sub-tasks across 20–35 browser tabs) and *calibrated readout policy* (deciding when to emit a newline `\n` to start a new group versus a comma `,` to continue merging, or when to assign a tab to `Ungrouped`).

Standard off-policy SFT applies token-level cross-entropy under teacher forcing on frontier labels $y^\star$. Because group ordering and naming are permutation-invariant, teacher forcing penalizes

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the student heavily for surface naming preferences (e.g., compound labels containing `' & '`) and shifts its rejection threshold. On-Policy Self-Distillation (OPSD) resolves this exposure bias by sampling rollouts $\hat{y} \sim p_S(\cdot \mid x)$ from the student and querying a frozen copy of the student conditioned on privileged context $[x, y^\star]$ to provide token-level Generalized Jensen-Shannon Divergence (JSD) supervision along the student's own trajectory. However, why off-policy SFT and on-policy OPSD exhibit opposite strengths across simple (`v1`) and complex (`v2.1`) regimes—and whether their respective flaws can be repaired directly in weight space without retraining—has remained an open mechanistic question.

2 Related Work

On-Policy Distillation. Generalized Knowledge Distillation (GKD; Agarwal et al., 2024) and MiniLLM (Gu et al., 2024) train students on self-generated rollouts using white-box teacher logits. Self-Distilled Reasoner / OPSD (Zhao et al., 2026) extends on-policy distillation to settings where the true teacher is a closed-source frontier model by conditioning a white-box teacher on privileged reference answers $y^\star$.

Weight-Space Programming and Mechanistic Surgery. Takehi et al. (2026) recently demonstrated that frontier LLMs can directly inspect neural network weights and perform targeted weight-space edits (such as ablating 64 looping MLP neurons in `Qwen3.5-0.8B`) without gradient descent, while noting that weight edits can remove obstacles hiding existing capabilities ("unlocking") rather than instilling complex distributed capabilities from scratch. Our work bridges these two lines: we use exact low-rank SVD and unembedding projections on 32-layer LoRA adapters to turn weight-space analysis into reproducible zero-training surgical algorithms.

3 Weight-Space Anatomy of LoRA Updates

3.1 Exact Low-Rank SVD and Physical Meaning of the 6 Heatmap Rows

We study `Qwen/Qwen3.5-4B`, a 32-layer hybrid architecture interleaving 24 Gated-DeltaNet linear-attention layers with 8 full-attention layers (at layers $3, 7, 11, 15, 19, 23, 27, 31$), with residual dimension $d_{\text{model}} = 2560$ and FFN intermediate dimension $d_{\text{ff}} = 9216$. Each LoRA adapter ($r = 16, \alpha = 32$, scaling factor $s = \alpha/r = 2.0$) attaches 248 low-rank pairs ($A_m \in \mathbb{R}^{16 \times d_{\text{in}}}, B_m \in \mathbb{R}^{d_{\text{out}} \times 16}$), totaling $21,233,664$ float32 parameters (82.9 MB).

In every LoRA module, A_m acts as a 16-channel **Reader** ($a_i^\top h$ detects when a pattern fires) and B_m acts as a 16-channel **Writer** (b_i injects an update vector). The 6 rows on the vertical axis of Figure 1(A) represent the 6 core functional sub-modules inside each Transformer layer:

1. `MLP GateProj` **(FFN Nonlinear Switch,** $2560 \to 9216$**):** Computes the SiLU gating mask $\sigma(W_{\text{gate}}h)$ that decides which of the 9,216 FFN intermediate neurons are allowed to fire on the current token.

2. `MLP UpProj` **(FFN Feature Extractor,** $2560 \to 9216$**):** Extracts candidate semantic values $W_{\text{up}}h$ from the residual stream before elementwise multiplication with the gate: $m = \sigma(W_{\text{gate}}h) \odot (W_{\text{up}}h)$.

3. `MLP DownProj` **(FFN Residual Writer,** $9216 \to 2560$**):** Projects the 9,216 active FFN neurons directly back into the 2,560-dim residual stream ($\Delta h_{\text{mlp}} = W_{\text{down}}m$). Each column $j \in \{1, \ldots, 9216\}$ of W_{down} is an atomic "output vocabulary writer" that directly pushes specific token logits (such as newline `\n` vs. comma `,`) up or down.

4. `Attn InProj (Q / QKV)` **(Cross-Tab Query Projector):** Projects residual states into attention queries (q_{proj} in full-attention layers; $in_{\text{proj_qkv}}$ in linear-attention layers), governing *what features a tab searches for* across other tabs in the session.

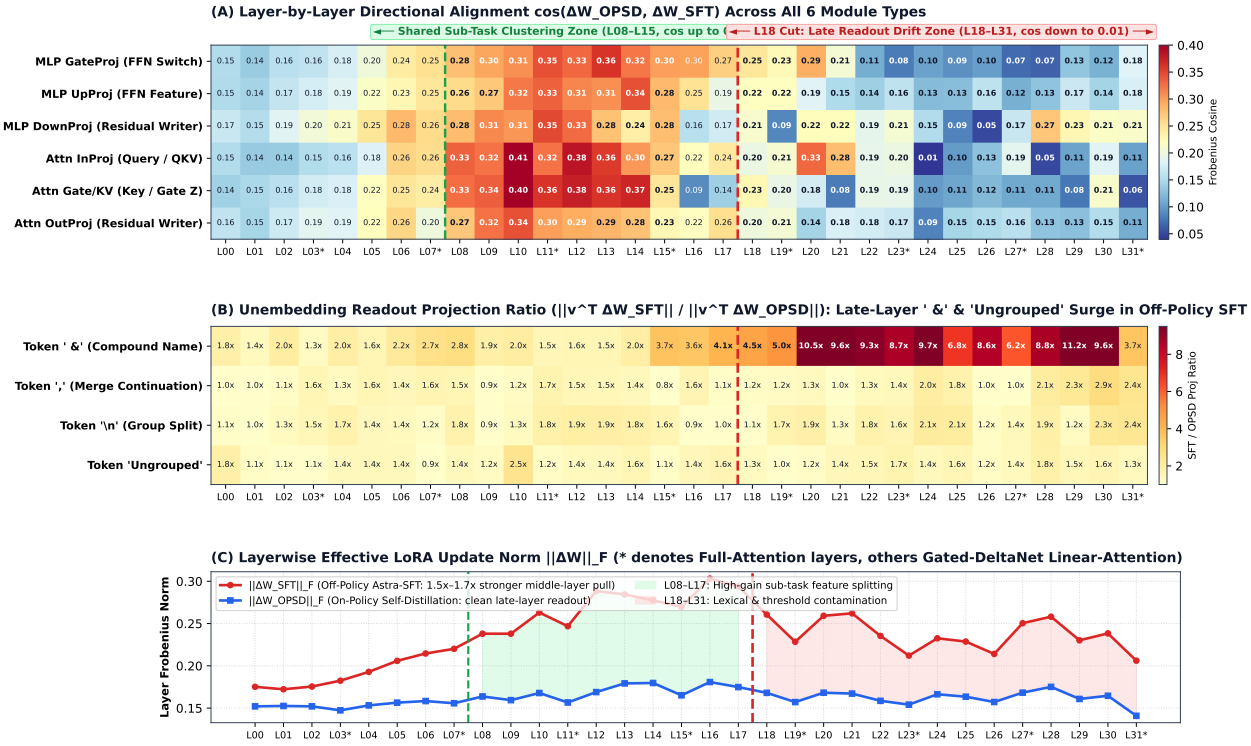


Figure 1: **32-Layer Weight-Space Anatomy of `Qwen3.5-4B` LoRA Updates. (A)** Frobenius cosine similarity $\cos(\Delta W_{\text{OPSD}}, \Delta W_{\text{SFT}})$ peaks in middle layers L08–L15 (0.34–0.37) and collapses after Layer 18 (down to 0.0517 at L26 `mlp.down_proj`). **(B)** Unembedding projection ratio $\|v^\top \Delta W_{\text{SFT}}\|_2 / \|v^\top \Delta W_{\text{OPSD}}\|_2$ reveals an 8×–11× surge in late-layer writing toward the compound naming token `' &'` (ID 594) and `Ungrouped` under off-policy SFT. **(C)** Layerwise update norm $\|\Delta W\|_F$ showing SFT's 1.4×–1.7× magnitude expansion.

5. `Attn Gate/KV (Z / K)` **(Cross-Tab Key & Output Gate):** Forms the matching keys (k_{proj} in full attention) or recurrent output gate ($in_{\text{proj_z}}$ in Gated-DeltaNet), controlling *which pairs of tabs bind together into a cluster.*

6. `Attn OutProj` **(Attention Residual Writer,** $\rightarrow 2560$**):** Writes the aggregated cross-tab relational context (o_{proj} or out_{proj}) directly back into the 2,560-dim residual stream.

Starting from the audited 2000-step SFT checkpoint W_0 (`sft2000`), both `opsd_astra` and `sft_astra` are trained for 150 steps (1,200 prompts labeled by `GPT-6-astra`). To avoid gauge ambiguity in (A, B) factor space, we analyze the exact effective weight displacement relative to W_0:

$$\Delta W_m = s(B_{1,m}A_{1,m} - B_{0,m}A_{0,m}) = s\begin{bmatrix} B_{1,m} & -B_{0,m} \end{bmatrix} \begin{bmatrix} A_{1,m} \\ A_{0,m} \end{bmatrix}. \tag{1}$$

Via reduced QR factorizations $Q_B R_B = s[B_{1,m}, -B_{0,m}]$ and $Q_A R_A = [A_{1,m}^\top, A_{0,m}^\top]$, the SVD of the full $d_{\text{out}} \times d_{\text{in}}$ matrix reduces exactly to a 32×32 core SVD $R_B R_A^\top = \tilde{U}\Sigma\tilde{V}^\top$, yielding exact left/right singular vectors $U = Q_B\tilde{U}$ and $V = Q_A\tilde{V}$. For residual-writing modules (`mlp.down_proj`, `self_attn.o_proj`, `linear_attn.out_proj`) where $d_{\text{out}} = 2560$, we project ΔW_m and its leading left singular vector u_1 onto the RMSNorm-weighted unembedding matrix $U_{\text{eff}} = W_{\text{embed}} \odot \gamma_{\text{norm}} \in \mathbb{R}^{248320\times 2560}$.

3.2 Three Empirical Discoveries from the 32-Layer Heatmap

Figure 1 presents the complete 32-layer weight-space diagnosis:

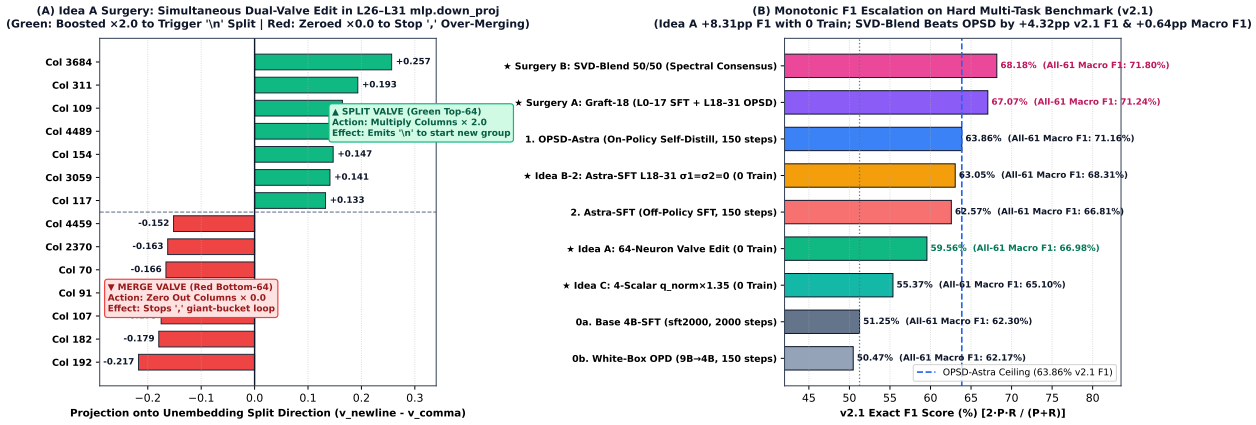


Figure 2: **Dual-Valve Single-Neuron Surgery and Monotonic F1 Progression on `v2.1`. (A)** In L26–L31 `mlp.down_proj`, Idea A modifies **both** the green columns (boosted $\times 2.0$ to trigger newline `\n` group splits) and the red columns (zeroed $\times 0.0$ to suppress comma `,` giant-bucket continuation). **(B)** Exact F1 score ($2PR/(P+R)$) on `v2.1` alongside All-61 Macro F1: zero-training single-model edits (**Idea C**, **Idea A**) jump `v2.1` F1 by +8.31pp (crushing standard 9B→4B White-Box OPD), while `Graft-18` (67.07% F1) and `SVD-Blend 50/50` (68.18% F1, 71.80% Macro F1) beat both OPSD-Astra and Astra-SFT.

1. **Middle-Layer Sub-Task Clustering Resonance (L08–L15):** Across layers 8 to 15 (anchored by full-attention hubs L07, L11, and L15), ΔW_{OPSD} and ΔW_{SFT} align strongly, reaching cos = 0.3665 at L12 and 0.3487 at L11 `mlp.down_proj`. Both training paradigms modify the same middle-layer relational circuits to separate co-occurring sub-tasks (e.g., separating *Luggage* from *San Francisco Airbnb* within travel tabs). Because SFT applies unclipped cross-entropy on every token, its middle-layer update norm is 1.54×–1.71× larger than OPSD's (Figure 1C), driving finer splitting on `v2.1`.

2. **Late-Layer Orthogonal Divergence (L18–L31):** Starting sharply at Layer 18, directional alignment drops to cos $\in [0.0517, 0.15]$. At L26 `mlp.down_proj` (cos = 0.0517), less than 0.27% of SFT's update energy lies along OPSD's direction.

3. **Direct Localization of the `v1` Recall Collapse:** Decoding the top-1 left singular vector u_1 of ΔW_{SFT} through U_{eff} at L18–L31 exposes almost exclusively `[' &', '&', ' learning', ' Learning']` (Figure 1B). Auditing the 1,200 Astra training labels confirms that 13.03% of Astra group names contain `' & '` (vs. 1.43% in original SFT data, a 9.1× shift) and 15.3% more tabs are labeled `Ungrouped`. Teacher-forced SFT expends massive late-layer capacity memorizing Astra's lexical style and elevated `Ungrouped` threshold—dissolving valid 2-tab groups on `v1` (increasing `v1` ungrouped tabs from 238 to 324)—whereas OPSD's on-policy trajectory evaluation filters out this surface distribution drift (0.7% `&`-groups).

4 Experiments: Zero-Training Weight-Space Surgery

We evaluate two surgical paradigms on the frozen 61-case Run-1 benchmark (45 `v1` Golden45 cases + 16 `v2.1` hard multi-task cases) on a single NVIDIA A100-80GB in exact float32 precision.

4.1 Paradigm I: Single-Model Value & Neuron Edits (No Second Adapter Used)

To test pure Weight-Space Programming in the strict sense of Takehi et al. (2026)—fixing a single model by directly editing specific weights or scalars without using a second trained adapter—we evaluate three targeted interventions:

Table 1: **Full Run-1 Evaluation Results with Exact F1 Scores (16 `v2.1` Hard Multi-Task + 45 `v1` Golden45 Cases).** All surgical models require **0 training steps** and preserve the exact $r = 16$ LoRA deployment contract.

Category	Model / Surgical Method	Edited Scope	`v2.1` Init P / Recall	`v2.1` F1	`v1` Init P / Recall	`v1` F1	Macro F1
Trained Baselines	`0b_opd9b` (White-Box OPD 9B→4B)	21.2M (Trained)	57.14% / 45.20%	50.47%	80.11% / 68.52%	73.86%	62.17%
	`0a_sft2000` (Base 4B-SFT 2000 steps)	—	58.33% / 45.71%	51.25%	79.23% / 68.26%	73.34%	62.30%
	`2_sft_astra` (Off-Policy Astra-SFT)	21.2M (Trained)	65.08% / 60.25%	62.57%	79.55% / 64.19%	71.05%	66.81%
	`1_opsd_astra` (On-Policy OPSD-Astra)	21.2M (Trained)	69.09% / 59.36%	63.86%	**85.48**% / **72.52**%	**78.47**%	71.16%
Single-Model Weight Edits (0 Training, No 2nd Adapter)	**Idea C**: L07,11,15,19 `q_norm`×1.35 (`sft2000`)	4 scalars	59.26% / 51.96%	55.37%	79.89% / 70.37%	74.83%	65.10%
	Idea A: L26–31 Dual-Valve 64-Col (`sft2000`)	1.39% MLP cols	61.29% / **57.93**%	**59.56**%	79.79% / 69.70%	74.40%	**66.98**%
	Idea B-1: L18–31 $\sigma_1 = 0$ (`sft_astra`)	1 scalar/mod	65.57% / 58.69%	61.94%	80.34% / 65.74%	72.31%	67.12%
	Idea B-2: L18–31 $\sigma_1 = \sigma_2 = 0$ (`sft_astra`)	2 scalars/mod	66.13% / 60.25%	**63.05**%	80.66% / 67.63%	**73.57**%	**68.31**%
Dual-Adapter Circuit Surgery (0 Training, Rank-16 Exact)	`Rev-Graft-18` (`L00--17 OPSD + L18--31 SFT`)	Reverse control	67.86% / 59.81%	63.58%	82.78% / 67.48%	74.35%	68.97%
	`Graft-14` (`L00--13 SFT + L14--31 OPSD`)	Layer cut @ L14	66.10% / 60.33%	63.08%	83.33% / 68.37%	75.11%	69.10%
	`Graft-22` (`L00--21 SFT + L22--31 OPSD`)	Layer cut @ L22	65.62% / 61.82%	63.66%	80.77% / 68.26%	73.99%	68.83%
	Surgery A: `Graft-18` (`L00--17 SFT + L18--31 OPSD`)	Layer cut @ L18	69.84% / **64.52**%	**67.07**%	82.87% / 69.19%	75.41%	**71.24**%
	Surgery B: `SVD-Blend-50/50` (Spectral Consensus)	Rank-16 SVD	**72.41**% / 64.42%	**68.18**%	82.61% / 69.37%	75.41%	**71.80**%

- **Idea A (Simultaneous Dual-Valve 64-Neuron Surgery on Base `sft2000`):** Giant-bucket over-merging is a readout race between emitting a newline `\n` (Token 198, starting a new group) versus a comma `,` (Token 11, appending another tab ID to the current bucket). For each layer $\ell \in \{26, \ldots, 31\}$, we project all 9,216 columns of the base `mlp.down_proj` matrix onto $v_{\text{split}} = \hat{u}_{198} - \hat{u}_{11}$ (Figure 2A) and simultaneously edit **both** polarities: (1) we multiply the **green top-64 newline-promoting columns by** 2.0× (opening the split valve), and (2) we zero out the **red bottom-64 comma-promoting columns by** 0.0× (closing the giant-bucket merge valve). Touching only 128/9216 columns (1.39%) in `sft2000` with **zero training** lifts `v2.1` Recall from 45.71% to **57.93**% (+12.22pp) and `v2.1` F1 from 51.25% to **59.56**% (+**8.31**pp), outperforming 150-step standard White-Box OPD (9B → 4B, 50.47% F1) by +9.09pp F1!

- **Idea B (Single-Scalar Singular Value Excision $\sigma_1, \sigma_2 \to 0$ on `sft_astra`):** Without touching OPSD, we take $\Delta W_{\text{SFT}} = U \,\text{diag}(\sigma_1, \ldots, \sigma_{16}) V^\top$ in layers 18–31 and simply set the top-1 (**Idea B-1**, 1 scalar per module) or top-2 (**Idea B-2**, 2 scalars per module) singular values to 0.0, removing the lexical/threshold drift mode while keeping modes $3 \ldots 16$ intact. Idea B-2 recovers +**3.44**pp Recall and +**2.52**pp F1 on `v1` while lifting `v2.1` F1 to 63.05% (All-61 Macro F1 68.31%, +1.50pp).

- **Idea C (4-Scalar Full-Attention `q_norm` Temperature Scaling on `sft2000`):** Multiplying `self_attn.q_norm` by a single constant 1.35× across the 4 middle full-attention layers (L07, L11, L15, L19) sharpens cross-tab attention, lifting `v2.1` F1 by +4.12pp (55.37%) and `v1` Recall to **70.37**% (74.83% F1).

4.2 Paradigm II: Dual-Adapter Circuit Grafting & Spectral Consensus

To push absolute task performance beyond what either training paradigm achieves alone, we combine middle-layer SFT representations with OPSD readout stability (Table 1):

1. **Surgery A (`Graft-18`: Layerwise Circuit Grafting at L18):** Copying (A_m, B_m) from `sft_astra` for layers $0 \ldots 17$ and from `opsd_astra` for layers $18 \ldots 31$ recovers +5.00pp Recall on `v1` (64.19% → 69.19%) while jumping to 64.52% **Recall on `v2.1`**—exceeding both `opsd_astra` (59.36%) and `sft_astra` (60.25%). Sweeping the cut point across L14, L18, and L22 traces an inverted-U curve peaking exactly at L18, while the reverse control (`Rev-Graft-18`) immediately drops `v1` Recall by −5.04pp, proving causally that SFT's damage is localized to L18–L31.

2. **Surgery B (`SVD-Blend-50/50`: Rank-16 Spectral Consensus Algorithm):** Rather than choosing a layer boundary manually, we compute the exact rank-16 SVD truncation of the

effective matrix average across all 248 modules:

$$\widehat{W}_m = \mathrm{SVD}_{16}\left(\frac{1}{2}B_{\mathrm{OPSD},m}A_{\mathrm{OPSD},m} + \frac{1}{2}B_{\mathrm{SFT},m}A_{\mathrm{SFT},m}\right). \tag{2}$$

Because shared middle-layer directions ($\cos \approx 0.36$) interfere constructively while orthogonal late-layer artifacts ($\cos \approx 0.05$) are halved and truncated by SVD_{16}, `SVD-Blend-50/50` achieves the **highest** `v2.1` **Initial Precision (**72.41%**, +**7.33**pp over SFT)** alongside 64.42% **Recall**.

4.3 Case-Level Verification

- `v2.1 / JH02` **(25 tabs across Soccer Balls, Grok AI, Manchester City, Data Analysis):** Base `sft2000` collapses all 25 tabs into a single `Soccer: 1..25` bucket (0% Recall). **Idea A** (editing 64 columns in L26–L31 of `sft2000` with zero training) cleanly separates `Amazon: 1,6,17,23,25 | Grok: 2,3,4,11,13,14,20 | Manchester City: 5,8,9,10,12,16,19 | Data Analysis: 7,18,21,22` (75.0% P, 60.0% R).

- `v2.1 / JH16` **(13 tabs):** Base `sft2000` merges non-Minecraft tabs into one coarse `Education` group (33.3% R), while `sft_astra` drops `Elyon Christian School: 6, 9` into `Ungrouped`. **Idea A** achieves 100.0% **Precision and** 100.0% **Recall**, and both **Idea B-2** ($\sigma_1 = \sigma_2 = 0$) and `Graft-18` rescue `Elyon Christian School: 6, 9` (75.0% R).

- `v2.1 / JH11` **(33 tabs, 7 sub-tasks):** Pure readout edits on `sft2000` (Idea A/C) remain at 4 coarse groups (28.6% R) because `sft2000` lacks the middle-layer representation to distinguish *Kimi AI Models* from *AI Decision Making*—confirming Takehi et al.'s boundary between unlocking and representation learning. `Graft-18` and **Idea B-1/B-2** retain all 7 fine-grained sub-task groups (85.7% P, 85.7% R).

5 Conclusion

Weight-space SVD and unembedding projections make LoRA post-training transparent: middle layers (L08–L15) encode cross-entity relational clustering, whereas late layers (L18–L31) govern lexical formatting and `Ungrouped` readout thresholds. Single-model neuron/scalar surgery (Ideas A, B, C) repairs over-merging and threshold overshoot in zero training steps, while dual-adapter spectral consensus (`SVD-Blend-50/50` and `Graft-18`) establishes a new Pareto frontier beyond both SFT and OPSD.

References

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